

DEREK F. DILEO

Android Software Engineer | Kotlin · Jetpack Compose · UI Architecture

Andover, MA | (781) 316-6719 | derekdileo@gmail.com

[LinkedIn](#) | [GitHub](#) | derekdotdev.github.io

SUMMARY

Android Software Engineer and UI architecture owner specializing in Kotlin, Jetpack Compose, MVI, and Clean Architecture. Lead the design and implementation of scalable UI patterns, reusable Material 3 components, and dependency-inverted application boundaries. Experienced with lifecycle-aware foreground services, Room persistence, JNI-backed C++ integration, and automated testing across unit, instrumented, and Compose UI layers. Proven record of modernizing production Android systems and simplifying complex hardware-integrated workflows.

EXPERIENCE

Tomahawk Robotics / AeroVironment

Software Engineer II, Applications | 08/2022 – Present

Kotlin, Java, Jetpack Compose, Material 3, MVI, Clean Architecture, Hilt, Room, Coroutines/Flow, JNI/C++, JUnit, AssertK, React, TypeScript, AWS

- **Serve as the Android UI architecture owner, leading modernization of the Android codebase and establishing architectural patterns and development practices across the team;** drove the transition from imperative Java, LiveData, and observer-based patterns to Jetpack Compose, Material 3, MVI, Clean Architecture, and Hilt, while introducing reusable themed components, layered architecture and automated testing practices to improve maintainability, testability, and scalability
- **Expanded automated testing across the Android stack;** introduced pure Kotlin unit tests, instrumented tests, and Jetpack Compose UI tests using JUnit and AssertK.
- **Architected and delivered turnkey vehicle-pairing applications that transformed a manual, CLI-driven integration process into an automated UI workflow,** dynamically discovering and configuring ground radios to vehicle-specific channels and settings as operators enable or switch between nodes, significantly reducing the complexity of integrating and operating multiple vehicle platforms.
- **Integrated JNI to bridge native C++ libraries with Android components,** enabling Android applications to leverage existing native capabilities.
- **Built and deployed a full-stack customer self-service portal** using React, TypeScript, and AWS GovCloud, replacing a manual hardware-registration and software-licensing process with secure, 24/7 customer activation; enabled pre-provisioned license entitlements and significantly reduced support FTE overhead.
- **Automated bug reporting and issue tracking** through Jira API integration, eliminating manual ticket creation and streamlining issue intake for customers and support teams.

City of Orlando – Unhoused Youth Solutions

Technical Lead & Full-Stack Developer | 04/2022 – 08/2022

Capstone Project | Java, Spring Boot, Spring Security, JPA/Hibernate, Thymeleaf, MySQL, AWS RDS

- **Led a development team in designing and delivering a full-stack application for the City of Orlando** to support identification and outreach efforts for unhoused youth; owned task delegation, code reviews, technical coordination, and team communication.
- **Worked directly with City of Orlando stakeholders to gather requirements** and translate municipal needs into actionable technical requirements and application features.

- **Designed and implemented the application's relational data model and persistence layer** using UML, MySQL, Spring Data JPA, Hibernate, and AWS RDS.

Rx Delivered Now

Quality Assurance Manager | 03/2022 – 08/2022

Flutter/Dart, Firebase, Google Analytics, Jira, Confluence

- **Led quality assurance for mobile and web applications**, partnering with engineering, product, and executive leadership to identify defects, improve user experience, and drive issues through resolution.
- **Implemented Google Tag Manager and analytics tooling** to support product analytics, conversion tracking, and marketing insights.

Valencia College – School of Arts & Entertainment | Orlando, FL

Lab Assistant II | Systems & Dante Network Administration | 09/2015 – 08/2021

Linux, Docker, VirtualBox, Dante Domain Manager, Cisco Networking, macOS, Windows

- **Expanded the role beyond traditional lab support by identifying and implementing improvements to campus production infrastructure**, ultimately taking ownership of systems administration, networking, technical documentation, and support across complex audio environments.
- **Designed and deployed what became the largest educational RedNet network in North America**, using Linux virtualization, Dockerized infrastructure, and proprietary networking tools to support isolated virtual audio environments.
- **Administered and supported production infrastructure** spanning Linux, macOS, Windows, Cisco networking, professional audio systems, and KVM hardware while supervising student lab assistants and documenting system configuration and troubleshooting procedures.

Chosen Project - [Request Live Music | 2021](#)

Creator & Full-Stack Developer

Conceived, designed, and developed a cross-platform Flutter application for iOS, Android, and web that allowed users to discover nearby entertainers and submit contact-free song requests. Implemented location-based discovery, push notifications through Firebase Cloud Messaging, and a dedicated request-management dashboard for performers.

TECHNICAL SKILLS

Android: Kotlin, Java, Jetpack Compose, Android Views/XML, MVI, MVVM, Clean Architecture, Coroutines, Flow, Room, Hilt, JNI

Testing & Development: JUnit, TDD, Gradle, Git, GitHub, GitLab, Jenkins, Jira, Postman

Backend & Cloud: AWS, AWS GovCloud, SQL, Spring Boot, REST APIs, Hibernate, JPA

Web & Cross-Platform: TypeScript, JavaScript, React, Flutter, Dart, Node.js

Infrastructure: Docker, Linux, networking, virtualization

AI-Assisted Development: Claude Code, ChatGPT

EDUCATION

Valencia College | Orlando, FL

B.A.S., Computer Engineering & Software Development | 2022

A.S., Audio Engineering | 2020

A.S., Sound Production | 2020

3.92 GPA · National Society of Collegiate Scholars · Multiple Academic Scholarships